Programmable Mathematical Execution for Digital Twins

An auxiliary runtime for explicit surrogate models and large parameter studies

Scientific working paper | Status: research proposal | 4 November 2026

Research opportunity

Digital twins become especially valuable when the same model is evaluated under many operating scenarios and updated measurements. Some model components are explicit algebraic functions that can be evaluated independently from the numerical integration of the system dynamics. paceval UG proposes a reusable mathematical layer for these components, using the Functional Mock-up Interface as an integration reference.



Figure 1. Proposed integration boundary for a reusable mathematical evaluation kernel.

1. Existing research tools and the exact gap to investigate

The Functional Mock-up Interface (FMI) specifies model exchange and co-simulation as distinct interfaces. In FMI Model Exchange, numerical time integration is carried out by the importing environment. This distinction is central to the proposed architecture. paceval would evaluate selected explicit expressions; integration, events, state bookkeeping and sensitivity computations remain with suitable simulation tools. [1,2]

2. Research contribution proposed by paceval

A standardized adapter to prepared mathematical graphs could simplify repeated parameter sweeps, calibration formulas and reduced physical model components. The main opportunity is a large collection of similar formula evaluations with changing parameters, rather than accelerating every type of differential-equation simulation.

paceval is a mathematical engine that transforms supported text expressions into reusable computation objects. Its published C/C++ API supports preparing calculations for repeated evaluation, including batched input sets. The initial study should use local library integration. Networked REST or MCP communication would distort latency measurements for the targeted arithmetic kernel. The product documentation is linked in the references.

3. Explicit mathematical test case

A one-state lumped thermal model is C dT/dt = P_in - hA(T - T_amb), with thermal capacitance C > 0 and heat-transfer conductance hA > 0. The algebraic right-hand side f(T,P_in,hA,T_amb,C) = [P_in - hA(T - T_amb)]/C is a suitable reusable expression. For constant inputs, the exact solution is T(t) = T_amb + P_in/(hA) + [T(0) - T_amb - P_in/(hA)] exp[-hA t/C].

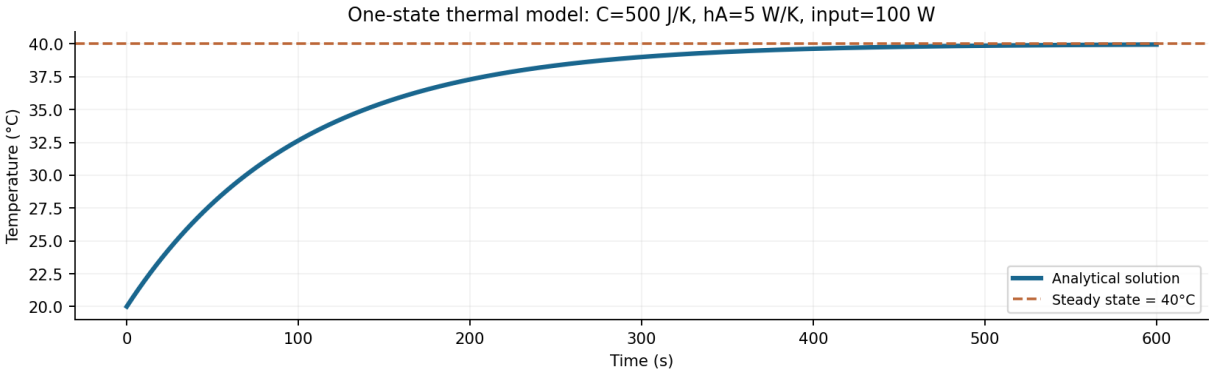


Figure 2. Exact temperature evolution of a one-state thermal model. Synthetic or analytic example, not a paceval performance measurement.

The graph uses specified synthetic physical parameters and the exact analytic solution of the one-state thermal differential equation. It is an illustration of a correctly separated model function and time integrator, not a paceval benchmark or experimental device measurement.

4. Reproducible pilot experiment

Begin with the one-state model and progress to a documented nonlinear surrogate. Implement identical right-hand sides using paceval and compiled C++ through a common interface. Use exactly the same ODE solver tolerances, event handling and initial states. Vary the number of scenarios, model parameters, function call counts and fractions of total runtime spent in the algebraic component.

Evaluation dimension	Study design
Mathematical correctness	Identical formulas and input domain. Independent numerical references, including boundary cases.
Kernel efficiency	Function calls per second; total scenario simulation time; agreement of time trajectories; model initialization; solver call counts; energy and memory per parameter sweep.
Reproducibility	Thirty or more repetitions after warm-up, published inputs and software versions. Report median, dispersion and uncertainty.
Full-system validity	Include graph creation, adapters, data movement and the effect on total execution time.

5. Integration with established scientific software

A local C-compatible wrapper may expose the selected right-hand-side functions to an FMI-compatible importing environment. This is a proposed adapter, not a claim that paceval currently implements the FMI standard. The host retains simulation time, integration algorithm, model events, units and state consistency.

6. Programmable FPGA and ASIC perspective

Future on-device digital twins may employ direct streaming from sensors into explicit surrogate functions. An FPGA path should be evaluated only after profiling shows a substantial arithmetic workload and includes transfer costs and energy per simulated scenario.

7. Expected scientific contribution

The work could establish a reusable method for evaluating explicit simulation submodels on different execution backends while preserving fidelity and interoperability with established numerical solvers.

8. Open invitation to collaborate

paceval UG invites digital-twin researchers, modeling tool providers and FMI users to jointly develop an open adapter and assess repeated mathematical function execution in realistic multi-scenario simulations.

paceval UG is prepared to make the existing mathematical technology available for jointly defined research and to discuss targeted API extensions, open benchmark artifacts and integration formats with the scientific partners. Publication rights, licensing and intellectual property would be agreed before a joint project begins.

9. Practical work packages

First, implement the thermal reference model with a single verified numerical integrator. Next, create an optional FMI-side mathematical evaluator and compare complete trajectories across parameter sets. Finally, extend the test to a documented nonlinear model and report energy and memory costs.

Scientific and technical sources

[1] FMI 3.0.2 downloads and specifications. Open source 1

[2] FMI 3.0 specification, model exchange and co-simulation. Open source 2

[3] CasADi mathematics and simulation interfaces. Open source 3

[4] paceval C/C++ API. Open source 4

Technical information about paceval: paceval.com

Research contact: info@paceval.com